

```
DROP TABLE IF EXISTS
bu_frd.model_features_ab_final_2022_2025;
```

```
CREATE TABLE
bu_frd.model_features_ab_final_2022_2025
USING PARQUET
AS
```

```
SELECT
    party_id,
    observation_date,
    observation_year,
    model_split,
    label_3m,
    cohort_role,

    /* =====
    MODEL A
    SPARSE BEHAVIOR
    ===== */

    deterioration_acceleration,
    adverse_acceleration,
    max_dpd_jump_recent_3m,

    /* =====
    ACTIVE ACCOUNT BEHAVIOR
    ===== */

    active_accounts_change_vs_baseline,

    CASE
        WHEN ABS(active_accounts_change_vs_baseline) > 0
        THEN 1
        ELSE 0
    END AS active_accounts_changed_flag,

    /* =====
    BALANCE
    ===== */

    COALESCE(
        balance_change_vs_baseline,
        0.0
    ) AS balance_change_model,
```

```
COALESCE(  
    log1p_abs_balance_change,  
    0.0  
) AS log1p_abs_balance_change_model,
```

```
CASE  
    WHEN balance_change_vs_baseline IS NOT NULL  
    THEN 1 ELSE 0  
END AS balance_change_available_flag,
```

```
balance_recent_months_available,  
balance_prior_months_available,
```

```
/* =====  
PAST DUE  
===== */
```

```
COALESCE(  
    past_due_change_vs_baseline,  
    0.0  
) AS past_due_change_model,
```

```
CASE  
    WHEN past_due_change_vs_baseline IS NOT NULL  
    THEN 1 ELSE 0  
END AS past_due_change_available_flag,
```

```
past_due_recent_months_available,  
past_due_prior_months_available,
```

```
/* =====  
DELINQUENCY  
===== */
```

```
COALESCE(  
    delinquency_share_change_vs_baseline,  
    0.0  
) AS delinquency_share_change_model,
```

```
CASE  
    WHEN delinquency_share_change_vs_baseline IS NOT NULL  
    THEN 1 ELSE 0  
END AS delinquency_change_available_flag,
```

```
delinquency_recent_months_available,  
delinquency_prior_months_available,
```

```
/* =====  
UTILIZATION  
Same availability pattern for avg + weighted  
===== */
```

```
COALESCE(  
    utilization_change_vs_baseline,  
    0.0  
) AS avg_utilization_change_model,
```

```
COALESCE(  
    weighted_utilization_change_vs_baseline,  
    0.0  
) AS weighted_utilization_change_model,
```

```
CASE  
    WHEN utilization_change_vs_baseline IS NOT NULL  
    THEN 1 ELSE 0  
END AS utilization_change_available_flag,
```

```
utilization_recent_months_available,  
utilization_prior_months_available,
```

```
/* =====  
MODEL B ADDITIONS  
SELECTED Z FEATURES  
  
Missing Z → 0  
availability flag keeps "missing" separate from  
an actual standardized change of zero.  
===== */
```

```
COALESCE(  
    active_accounts_z_win,  
    0.0  
) AS active_accounts_z_model,  
  
active_accounts_z_available_flag,
```

```
COALESCE(  
    avg_utilization_z_win,  
    0.0  
) AS avg_utilization_z_model,
```

avg_utilization_z_available_flag,

COALESCE(

weighted_utilization_z_win,

0.0

) AS weighted_utilization_z_model,

weighted_utilization_z_available_flag

FROM

bu_frd.model_behavior_features_transformed_2022_2025;

SELECT

model_split,

label_3m,

COUNT(*) AS rows,

COUNT(DISTINCT party_id) AS parties,

SUM(

CASE

WHEN balance_change_model IS NULL

OR past_due_change_model IS NULL

OR delinquency_share_change_model IS NULL

OR avg_utilization_change_model IS NULL

OR weighted_utilization_change_model IS NULL

OR active_accounts_z_model IS NULL

OR avg_utilization_z_model IS NULL

OR weighted_utilization_z_model IS NULL

THEN 1 ELSE 0

END

) AS rows_with_model_feature_null

FROM bu_frd.model_features_ab_final_2022_2025

GROUP BY

model_split,

label_3m

ORDER BY

model_split,

label_3m DESC;